

13 Cardinality of Sets

13.1 Sets with Equal Cardinalities

A. Show that the two given sets have equal cardinality by describing a bijection from one to the other. Describe your bijection with a formula (not as a table).

2. $\mathbb{R}$ and $(\sqrt{2}, \infty)$

$$f(x) = e^x + \sqrt{2}, \text{ with inverse } f^{-1}(y) = \ln(y - \sqrt{2}).$$

4. The set of even integers and the set of odd integers

$$f(n) = n + 1, \text{ with inverse } f^{-1}(m) = m - 1.$$

6. $\mathbb{N}$ and $S = \{\frac{\sqrt{2}}{n} : n \in \mathbb{N}\}$

$$f(n) = \frac{\sqrt{2}}{n}. \text{ It is surjective by the definition of } S, \text{ and injective since } \frac{\sqrt{2}}{n} = \frac{\sqrt{2}}{m} \text{ gives } n = m.$$

8. $\mathbb{Z}$ and $S = \{x \in \mathbb{R} : \sin x = 1\}$

$$\text{Since } \sin x = 1 \text{ exactly when } x = \frac{\pi}{2} + 2\pi k \text{ for some } k \in \mathbb{Z}, \text{ take } f(k) = \frac{\pi}{2} + 2\pi k, \text{ with inverse } f^{-1}(x) = \frac{x - \frac{\pi}{2}}{2\pi}.$$

10. $\{0, 1\} \times \mathbb{N}$ and $\mathbb{Z}$

$$f(a, b) = a - 2ab + b, \text{ shown bijective in Exercise 12 of Section 12.2.}$$

12. $\mathbb{N}$ and $\mathbb{Z}$ (Suggestion: use Exercise 18 of Section 12.2.)

$$f(n) = \frac{(-1)^n(2n-1)+1}{4}, \text{ shown bijective in Exercise 18 of Section 12.2.}$$

14. $\mathbb{N} \times \mathbb{N}$ and $\{(n, m) \in \mathbb{N} \times \mathbb{N} : n \leq m\}$

$$f(n, m) = (n, n + m - 1), \text{ with inverse } f^{-1}(a, b) = (a, b - a + 1).$$

B. Answer the following questions concerning bijections from this section.

16. Verify that the function f in Example 13.3 is a bijection.

$$\text{Here } f : (0, \infty) \rightarrow (0, 1) \text{ is given by } f(x) = \frac{x}{x+1}.$$

It is injective: $\frac{x}{x+1} = \frac{y}{y+1}$ gives $x(y+1) = y(x+1)$, that is, $xy + x = xy + y$, so $x = y$.

It is surjective: given $y \in (0, 1)$, the number $x = \frac{y}{1-y}$ lies in $(0, \infty)$ and satisfies $f(x) = y$.

13.2 Countable and Uncountable Sets

2. Prove that the set $A = \{(m, n) \in \mathbb{N} \times \mathbb{N} : m \leq n\}$ is countably infinite.

4. Prove that the set of all irrational numbers is uncountable. (Suggestion: Consider proof by contradiction using Theorems 13.4 and 13.6.)

6. Prove or disprove: There exists a bijective function $f : \mathbb{Q} \rightarrow \mathbb{R}$.

8. Prove or disprove: The set $\mathbb{Z} \times \mathbb{Q}$ is countably infinite.

10. Prove or disprove: The set $A = \{\frac{\sqrt{2}}{n} : n \in \mathbb{N}\}$ is countably infinite.

12. Describe a partition of $\mathbb{N}$ that divides $\mathbb{N}$ into $\aleph_0$ countably infinite subsets.

14. Suppose $A = \{(m, n) \in \mathbb{N} \times \mathbb{R} : n = \pi m\}$. Is it true that $|\mathbb{N}| = |A|$?

13.3 Comparing Cardinalities

2. Prove that the set $\mathbb{C}$ of complex numbers is uncountable.
4. Prove or disprove: If $A \subseteq B \subseteq C$ and A and C are countably infinite, then B is countably infinite.
6. Prove or disprove: Every infinite set is a subset of a countably infinite set.
8. Prove or disprove: The set $\{(a_1, a_2, a_3, \dots) : a_i \in \mathbb{Z}\}$ of infinite sequences of integers is countably infinite.
10. Prove that if A and B are finite sets with $|A| = |B|$, then any surjection $f : A \rightarrow B$ is also an injection. Show this is not necessarily true if A and B are not finite.

13.4 The Cantor-Bernstein-Schröder Theorem

2. Show that $|\mathbb{R}^2| = |\mathbb{R}|$. Suggestion: Begin by showing $|(0, 1) \times (0, 1)| = |(0, 1)|$.
4. Let F be the set of all functions $\mathbb{R} \rightarrow \{0, 1\}$. Show that $|\mathbb{R}| < |F|$.
6. Show that $|\mathcal{P}(\mathbb{N} \times \mathbb{N})| = |\mathcal{P}(\mathbb{N})|$.